

What prompted you to start CynLr?

Nikhil: Gokul and I have known each other for about a decade now. We're both electronics and instrumentation engineers and we both were working in the same team at National Instruments (NI). NI is a tech giant headquartered in Austin, Texas. The company makes test and measurement equipment as well as automation hardware and software. One of NI's product lines is machine vision, for which they made smart cameras and software libraries that customers—especially from industrial scenarios (like manufacturing plants)—could use to put together a variety of machine vision applications.

Machine vision, at that point in time—and even now—is largely synonymous to inspection. People use cameras to identify a scene or an object in the scene and then make some decisions based on that, mostly for quality control purposes. For example, scanning a bar code is a machine vision application. There were many companies that focused on machine vision, including NI and Cognex. Later, as I took a sales role at NI, I got great exposure to what it takes to build large manufacturing plants and large industrial setups. Gokul specialised in machine vision. On the technical side, he ended up becoming the only machine vision specialist for a fairly large geographic area for NI.

We realised that the existing machine vision systems were only successful for simple identification problems. Machine vision was actually notoriously very difficult to solve. If customers had ten requirements, the existing systems at that time would succeed in only three out of the ten use cases. These systems were not meant for industrial robotics applications. They could not be used by robots to correctly grasp objects, pick, re-orient and manipulate them, and make proper placements in an industrial environ-

ment. Imagine a pick-and-place robot that's not able to properly grasp an object it's supposed to pick!

Machine vision for object manipulation had to be treated differently. The paradigms, algorithms, and architectures that were developed in machine vision, generally, were all for identification. People were trying to force-feed those algorithms for manipulation applications, and that was not working. That's when we started figuring out how we were going to solve this problem of vision for object manipulation.

SIGHT GIVES YOU A CERTAIN AMOUNT OF DATA ON A 3D OBJECT, BUT VISION GIVES MEANING TO THAT DATA.

In 2015, we left our jobs and approached customers we knew had a problem to be solved. Over the next few years we ended up solving over thirty unsolved machine vision solutions that involved some kind of manipulation!

Over time, we have achieved a certain level of standardisation. We raised a seed round of funding, and with that funding we built the hardware infrastructure. We also worked on the customer pipeline and built a team. And earlier this year, we closed our Pre-Series A to commercialise and take this technology to market. So, this has been our journey so far.

Could you elaborate on how hardware and software are looked at in your company?

Nikhil: One of the taglines that we write on our walls is, "Hardware is a part of the algorithm." On the vision side, especially, we

have built our own hardware. In fact, at a hardware level, we are the only depth perception system in the world to have eye movements, convergence, divergence, precision control, autofocus, etc that's enabled through hardware.

Gokul: Software cannot do magic. It's physical elements and physical objects that we're actually dealing with. So, without structuring the system to be able to pick or capture information about an object, we cannot create a holistic model of the object or intelligence about the object. Also, information about an object cannot be gotten just from a camera.

For example, we think autofocus just brings objects into focus, but actually it is the first layer of depth construction that we do. There are a lot of missing features in the way we use these physical devices like cameras. So, our inference algorithm begins with the hardware itself, it doesn't start with the software layers.

What is the most unique feature of your technology?

Gokul: Today, human labour is a straightforward process, because the same person with the same infrastructure, or without any infrastructure and additional investment, can go and work in a garment factory and stitch clothes. The next day, he can work at a different factory to put parts together to make an engine. On the third day, he can go into construction and start putting bricks together. He uses the same hands, eyes, and brain to handle all of these objects.

In the industry today, if you want to flip an object from one position to another (like from vertical to horizontal), you need to have an insane amount of infrastructure—around 30 to 40 different technologies are needed. That amounts to a lot of money. It also doesn't mean that the next day, I can reuse all of that infrastructure. What if I want to